

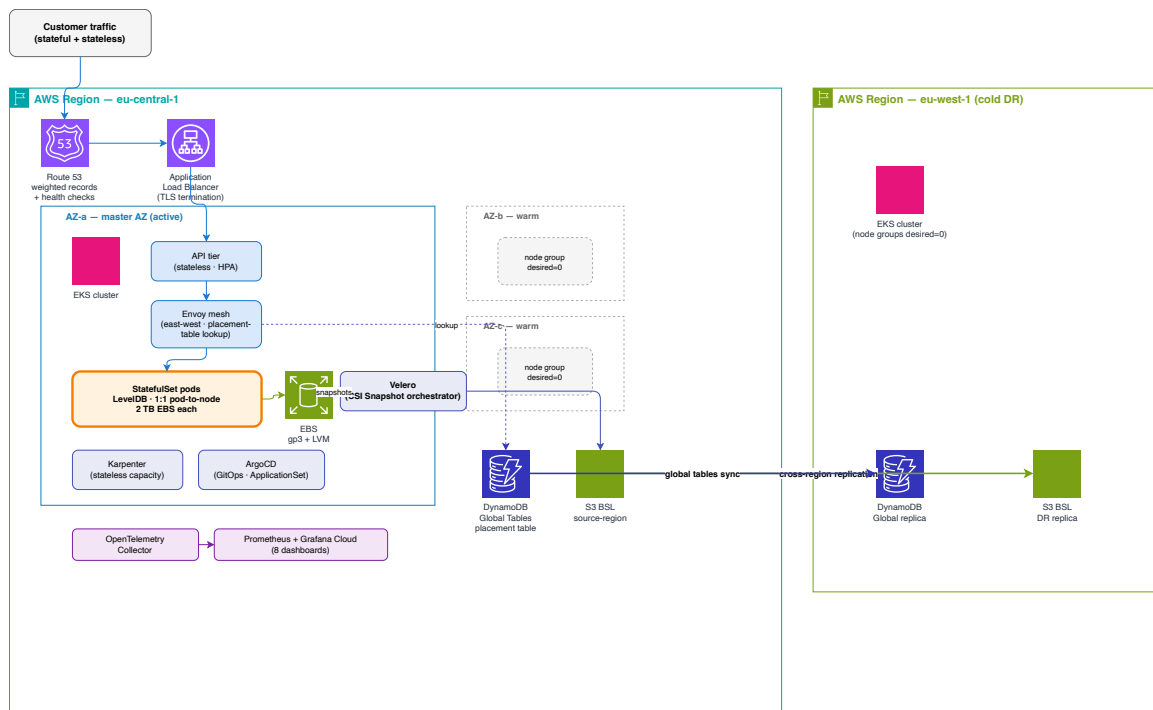
# aegis-statefulset — Architecture Overview

## aegis-statefulset

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## aegis-statefulset

Stateful Kubernetes workload management — proof-of-concept reference architecture for LevelDB-backed applications. Single-master-AZ topology, cold DR via Velero, Strangler-Fig migration support. Anonymised; portable across organisations.



# What this is

A parameterised reference architecture for hosting stateful, LevelDB-backed applications on Amazon EKS at Mittelstand-scale B2B SaaS. Built around three constraints: customer data on EBS is the only irreplaceable asset; operational overhead must be carriable by a small SRE team; cost must scale linearly with cell count (~\$2,700/month at POC `cells.count=1`). The architecture stays invariant across customer profiles — only `values.yaml` configuration moves.

## Who is this for

| You want to...                  | Start here                                                                      |
|---------------------------------|---------------------------------------------------------------------------------|
| Review the take-home submission | <a href="#">docs/SUBMISSION.md</a>                                              |
| Understand the architecture     | <a href="#">docs/architecture-overview.md</a>                                   |
| Read every decision             | <a href="#">10 thematic ADRs</a>                                                |
| Operate the platform            | <a href="#">Operations runbooks</a> + <a href="#">8 default dashboards</a>      |
| Quick-start a deployment        | <a href="#">Quick start</a> below                                               |
| Adapt to your context           | <a href="#">Configuration knobs</a> + <a href="#">values.yaml</a>               |
| Plan the long-term evolution    | <a href="#">Future-plans hub</a> — 5-horizon roadmap, triggers, cost trajectory |

## Quick start

```
# Prerequisites
#   aws CLI configured, terraform >= 1.6, helm >= 3.13, kubectl >= 1.28

# 1a. One-time per AWS account: provision the terraform state backend
#   (S3 bucket with native locking). See
#       infrastructure/terraform/bootstrap/README.md
cd infrastructure/terraform/bootstrap
terraform init && terraform apply
terraform output -raw backend_hcl_template > ../backend.hcl
cd ..

# 1b. Provision AWS infrastructure (main composition)
terraform init -backend-config=backend.hcl
terraform plan -out=tfplan && terraform apply tfplan

# 2. Configure kubeconfig
aws eks update-kubeconfig --name aegis-statefulset-prod --region eu-central-1

# 3. Bootstrap ArgoCD (one-time)
kubectl apply -n argocd -f gitops/argocd/projects/
```

```
kubectl apply -n argocd -f gitops/argocd/applications/observability-  
stack.yaml
```

#### # 4. Deploy applications (dev environment)

```
kubectl apply -n argocd -f gitops/argocd/applications/aegis-  
statefulset-dev.yaml
```

#### # 5. Verify

```
kubectl get statefulset -n aegis-app-dev  
helm test aegis-statefulset -n aegis-app-dev
```

Production sync is **manual** by design — see [gitops/argocd/README.md](https://github.com/gitops/argocd/blob/master/README.md) and ADR-08.

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## Architecture at a glance

Two organising principles plus one reliability discipline anchor every decision.

### P1 — Data on EBS is the only irreplaceable asset; everything else is declarative.

Customer state lives on encrypted EBS volumes — split per-pod into data and wal PVCs (Path γ multi-PVC; literal LVM available as opt-in patch per ADR-02). Cluster, deployments, routing tables, container images, AZ topology — all reproducible in 30–60 minutes from Helm + Terraform.

**P2 — RPO is a configurable trade-off, not a fixed floor.** Default 5-min cadence yields typical recovery point ~30 sec; the lever is exposed in `values.yaml`; the cost-RPO curve is documented for operator self-tune.

**P3 — Conservative on recovery, aggressive on failure detection.** Detection 50% unhealthy / 2 min sustained; recovery 95% healthy / 60 min sustained; asymmetric thresholds prevent flap. No auto-failback ever.

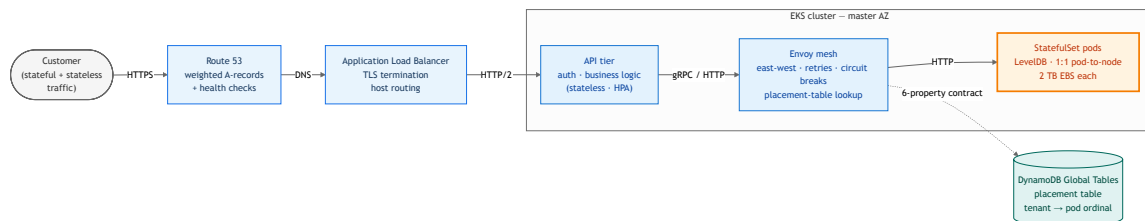
## Key choices

- **Per-tenant pod identity** — LevelDB is single-writer, file-based, no native replication; hash-sharding doesn't fit the storage primitive <sup>1</sup>
- **Single master AZ + warm-standby AZs** — cross-AZ chatter on a stateful tier without replication is wasted latency and double cost; AZ-B/C node groups sit at `desired=0` until a rotation event <sup>2</sup>
- **Multi-PVC storage (data + WAL) on EBS gp3** — the spec named LVM; modern K8s database operators (TiDB / K8ssandra / CloudNativePG) deliver the same WAL/data IO isolation + crash-consistent snapshot benefits via two PVCs + CSI VolumeSnapshot, without privileged init container or Kyverno PolicyException; literal LVM available as opt-in patch <sup>3 4</sup>
- **Velero CSI Snapshot path** — the spec named Restic + LVM; Velero still wraps Restic / Kopia in its File System Backup (FSB) path, so the transport changed, not the toolchain <sup>5</sup>
- **Cold DR over multi-region active-passive** — LevelDB has no sync APIs, so warm replicas are snapshot copies; the cost delta buys RTO that doesn't fit the ~25-min target <sup>6</sup>

- **DynamoDB Global Tables placement table** — 6-property contract (atomic compare-and-swap (CAS) / strong reads / multi-AZ durable / low latency / change-data-capture (CDC) / audit log); customer can substitute any backend that satisfies it <sup>7</sup>
  - **Strangler-Fig migration at the infrastructure layer** — application container untouched; cell-by-cell cutover via placement-table flip <sup>8</sup>
  - **Three-Layer DR** — Terraform infra / Helm via terraform helm\_release for cluster controllers / Velero for application + data; each layer the right tool, no double management <sup>9</sup>
  - **OpenTelemetry (OTel)-only instrumentation** — vendor-neutral at the instrumentation layer; backend (Grafana Cloud or Amazon Managed Prometheus + Grafana) reversible <sup>10</sup>
  - **Defense in depth** — Edge TLS → NetworkPolicy → Pod Security Standards (PSS) → Cosign admission → External Secrets Operator (ESO) + IAM Roles for Service Accounts (IRSA) → per-tier KMS → eBPF runtime → tamper-protected audit <sup>11</sup>
  - **SHA pin + Supply-chain Levels for Software Artifacts (SLSA) L3 + Software Bill of Materials (SBOM)** — five lines of YAML buy audit-grade chain of custody <sup>12</sup>
  - **Cost-as-architecture** — tagging discipline + AWS Budgets + Cost Anomaly Detector + per-tenant Cost and Usage Report (CUR) + Athena attribution + Savings Plans strategy; day-1, not bolted on <sup>13</sup>
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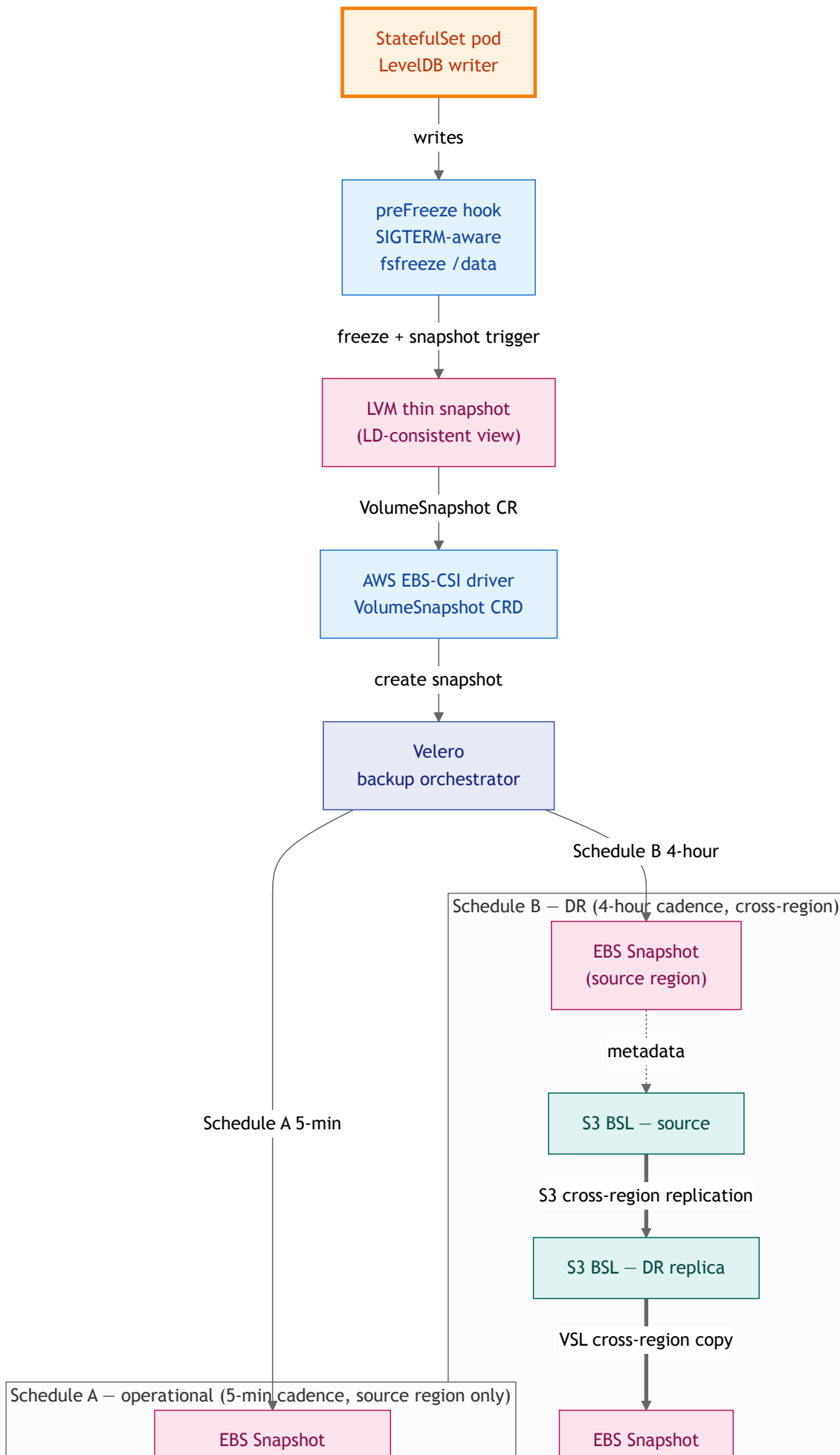
## Visual tour

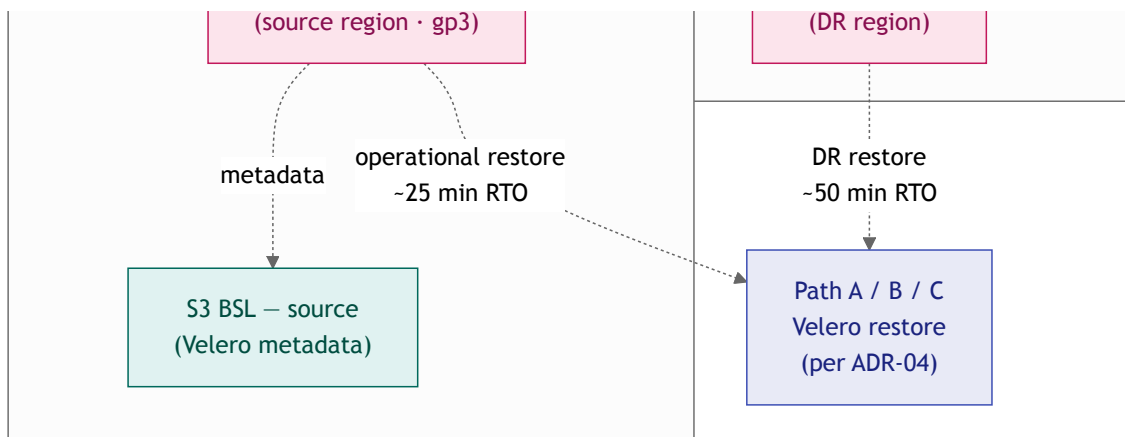
### Request flow — ALB → API → Envoy → StatefulSet



Edge / business-logic / routing / state are decoupled tiers. Each tier scales on its own shape — API on QPS, Envoy on routing-table churn, StatefulSet on tenant count. Pattern recognition, not invention: Vitess vtgate, DynamoDB request router, Stripe and Reddit shard-routers all do the same separation <sup>14</sup>.

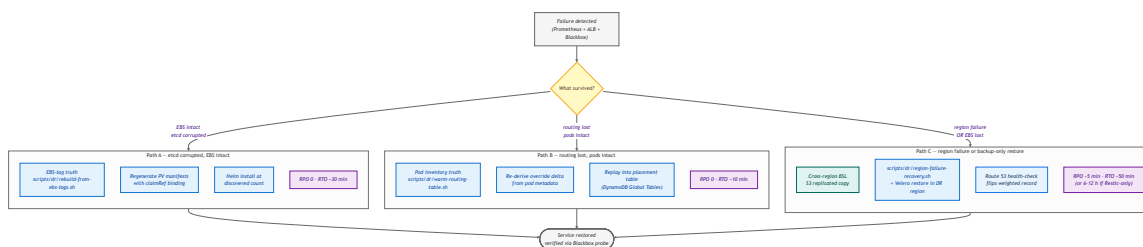
## **Backup data flow — Velero CSI Snapshot per PVC**





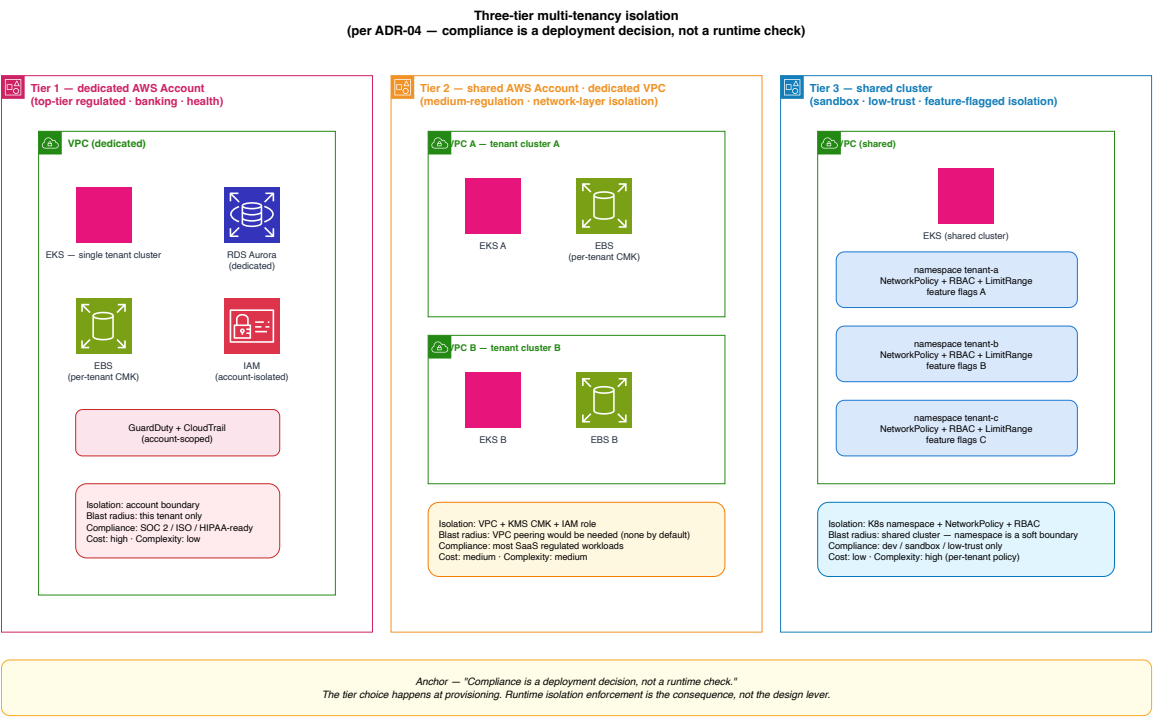
Velero pre-hook quiesces the app, then CSI VolumeSnapshot fires per PVC (both data and wal snapshot atomically), then EBS Snapshot lands in S3, then cross-region copy ships to Glacier IR. Dual-cadence schedules: Schedule A (5-min, source-region only) for operational restore; Schedule B (4-h, cross-region) for DR-tier <sup>15</sup> <sub>16</sub>.

## DR cutover — three paths based on what survived



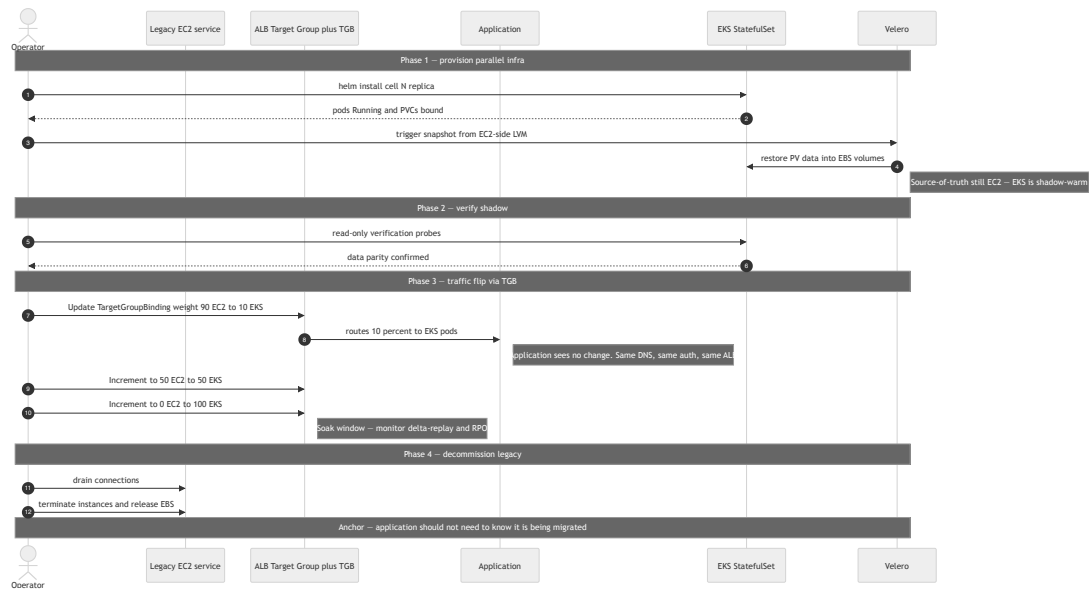
Recovery shape depends on the failure shape. Path A (EBS intact) is fastest; Path B (stateless tiers gone, EBS intact) means redeploy via Helm + reattach; Path C (region failure) means cross-region Velero restore. Each path has its own runbook + bounded RTO target <sup>17</sup> <sub>18</sub>.

# Multi-tenancy isolation — three tiers



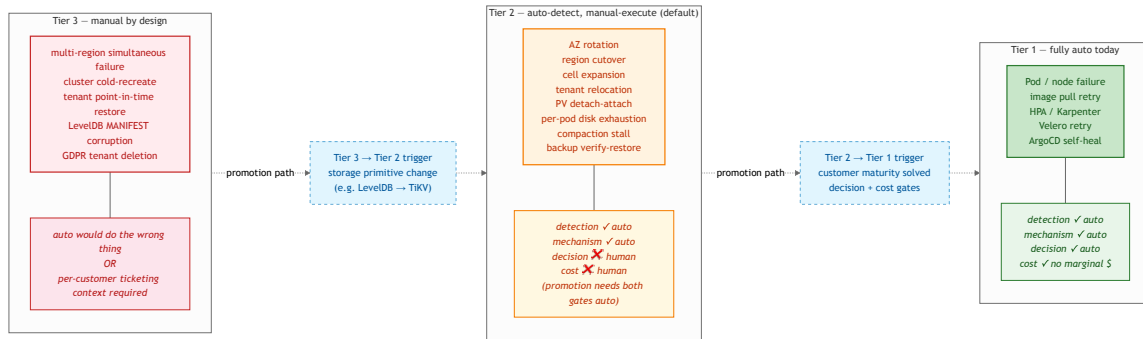
Three deployment-time tiers map cleanly to compliance posture: dedicated account / dedicated VPC / shared cluster + namespace. Runtime isolation enforcement (NetworkPolicy, IRSA, per-tier KMS) is the consequence, not the choice 19 20.

## Migration — Strangler Fig at the infrastructure layer



Application container untouched. Migration substrate change happens at the routing layer via TargetGroupBinding weights — 90/10, 50/50, 0/100, with soak-window monitoring at each step. The anchor: "the application should not need to know it's being migrated" 21.

## Automation tier matrix — failure-mode catalog



Every named failure mode is classified into one of three columns: auto-recovered / semi-auto (runbook-driven) / manual operator. The principle from P3 — detection automatic, execution manual — maps directly to this matrix. Optional opt-in patches in docs/future/ upgrade specific cells from semi-auto to auto <sup>22</sup>.

## Configuration

The operationally consequential knobs are summarised below; full schema lives in [helm/aegis-statefulset/values.yaml](#).

| Knob                                     | Default               | Trade-off                                                                  |
|------------------------------------------|-----------------------|----------------------------------------------------------------------------|
| backup.operational.cadence_minutes       | 5                     | Lower = tighter RPO + higher EBS Snapshot / S3 cost; spec ceiling 6 h      |
| backup.dr.cadence_hours                  | 4                     | DR-tier cadence; ≤ 5 h is the sane upper bound against the 6 h RPO ceiling |
| stateful.cells.count                     | 1                     | POC default 1; production matches existing partitioning per ADR-05         |
| stateful.startup_probe.failure_threshold | 60                    | 60 × 30s = 30 min cold-start window for production LDE MemTable rebuild    |
| stateful.cells.storage.data / .wal       | 1843Gi / 205Gi (prod) | Multi-PVC split (Path γ per                                                |

| Knob                  | Default       | Trade-off                                                                       |
|-----------------------|---------------|---------------------------------------------------------------------------------|
|                       |               | ADR-02); both PVCs CSI-snapshotted atomically; online-expandable via CSI resize |
| master_az             | eu-central-1a | Master AZ for all workloads; rotation via aws eks update-nodegroup-config       |
| dr_region             | eu-west-1     | Cross-region DR target for EBS Snapshot copy                                    |
| routing.backend       | dynamodb      | Customer can substitute any backend that satisfies the 6-property contract      |
| observability.backend | grafana_cloud | grafana_cloud or amp_amg — OTel-instrumented, backend reversible                |

## Repository layout

```

aegis-statefulset/
├─ helm/aegis-statefulset/      # Helm chart (templates, values,
environment overlays)
├─ infrastructure/terraform/    # AWS layer (VPC, EKS, IAM, S3, KMS,
FinOps, audit)
├─ scripts/                    # Backup / DR / migration / chaos /
FinOps
├─ gitops/                    # ArgoCD apps, Kyverno policies,
Grafana dashboards
├─ docs/                      # 10 thematic ADRs, ops runbooks,
architecture overview
├─ app/                       # POC mock binary (~80 lines Go; not
real LevelDB)
└─ .github/workflows/         # 4 main + 4 supporting CI workflows

```

Full per-file breakdown: [docs/architecture-overview.md](#).

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## Operational overhead awareness

The architecture ships maintenance surface — and acknowledges it on its own terms. Three tools to operate (Velero / EBS Snapshot / ArgoCD), six runbooks for human-side operations, ten thematic ADRs (originally split across fifty private predecessors), pre-commit's five-check enforcement gate, and a CI surface of four primary workflows plus four supporting. This is the senior platform-engineering job in audit-grade SaaS — not a side-effect to apologise for. Full breakdown plus the small-platform-team carrying-capacity argument: [docs/SUBMISSION.md § 7a](#).

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## Verification

```
helm lint helm/aegis-statefulset/  
helm template helm/aegis-statefulset/ | kubeconform -strict -summary  
cd infrastructure/terraform && terraform fmt -check -recursive &&  
    terraform validate  
shellcheck scripts/**/*.sh  
./scripts/git-hooks/pre-commit
```

Full local validation matrix: [CONTRIBUTING.md](#).

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## Status

Proof-of-concept submission for a take-home challenge. Architecture is stable; implementation is skeletal by design — the submission covers structural depth (10 thematic ADRs across 14 domains, parameterised topology, FinOps + GitOps + DevSecOps three-pillar discipline) over surface coverage. Anonymised for portability; reusable as a reference architecture for stateful Kubernetes workloads at Mittelstand-scale B2B SaaS.

## License

MIT — see [LICENSE](#).

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1. [docs/adr/ADR-01-architecture-and-topology.md](#)↩
2. [docs/adr/ADR-01-architecture-and-topology.md](#)↩
3. [docs/adr/ADR-02-storage-and-pv-mapping.md](#)↩
4. [docs/future/lvm-init-patch.md](#) — opt-in literal-LVM init container patch↩
5. [docs/operations/why-velero-not-restic.md](#)↩

6. [docs/operations/why-cold-dr.md↵](#)
7. [docs/adr/ADR-03-routing-and-ingress.md↵](#)
8. [docs/adr/ADR-05-migration-strangler-fig.md↵](#)
9. [docs/adr/ADR-04-backup-dr-and-ha.md↵](#)
10. [docs/adr/ADR-06-observability.md↵](#)
11. [docs/adr/ADR-07-security-and-runtime.md↵](#)
12. [docs/adr/ADR-09-supply-chain.md↵](#)
13. [docs/adr/ADR-10-finops.md↵](#)
14. [docs/adr/ADR-03-routing-and-ingress.md↵](#)
15. [docs/adr/ADR-04-backup-dr-and-ha.md↵](#)
16. [docs/operations/why-velero-not-restic.md↵](#)
17. [docs/operations/why-cold-dr.md↵](#)
18. [docs/adr/ADR-04-backup-dr-and-ha.md↵](#)
19. [docs/adr/ADR-01-architecture-and-topology.md↵](#)
20. [docs/adr/ADR-07-security-and-runtime.md↵](#)
21. [docs/adr/ADR-05-migration-strangler-fig.md↵](#)
22. [docs/adr/ADR-04-backup-dr-and-ha.md↵](#)